



Name: \_\_\_\_\_ Cohort/Batch: \_\_\_\_\_ Date: \_\_\_\_\_ Score: \_\_\_\_\_

## CLOUDFORMATION PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Cloud

TOTAL: 10 QUESTIONS

**Q1** What is CloudFormation and what is its primary purpose in its cloud ecosystem?

---

---

**Q6** How do you manage configuration and secrets for CloudFormation?

---

---

**Q2** How does IAM work with CloudFormation?

---

---

**Q7** What are the main cost drivers for CloudFormation and how do you optimize them?

---

---

**Q3** How do you monitor CloudFormation in production?

---

---

**Q8** How do you implement high availability with CloudFormation?

---

---

**Q4** How do you scale CloudFormation to handle variable load?

---

---

**Q9** How does CloudFormation integrate with a CI/CD pipeline?

---

---

**Q5** How do you secure CloudFormation in production?

---

---

**Q10** What are common pitfalls when first adopting CloudFormation?

---

---



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

**A1 CloudFormation is a managed cloud service that** Mitigation  
CloudFormation is a managed cloud service that handles a specific infrastructure concern (compute, storage, messaging, AI). Using it shifts operational burden to the cloud provider while you focus on application logic.

**A6 Store sensitive values in a managed secret** Observability  
Store sensitive values in a managed secret store (AWS Secrets Manager, GCP Secret Manager, Azure Key Vault). Inject secrets as environment variables at runtime, never bake them into container images or source code.

**A2 Access to CloudFormation is controlled via IAM** Primitives  
Access to CloudFormation is controlled via IAM roles and policies. Attach the minimum required permissions to a service account or role. Avoid long-lived access keys; use instance roles or Workload Identity Federation instead.

**A7 Cost in CloudFormation typically comes from compute, storage, and request counts.** Automation  
Cost in CloudFormation typically comes from compute hours, data transfer, storage, and request counts. Right-size instances, use reserved capacity for steady-state workloads, and enable lifecycle policies to expire old data.

**A3 CloudFormation emits metrics to the cloud provider's monitoring service** Hardening  
CloudFormation emits metrics to the cloud provider's monitoring service (CloudWatch, Cloud Monitoring, Azure Monitor). Set alerts on key metrics (error rate, latency, capacity). Use distributed tracing to correlate across services.

**A8 Deploy CloudFormation across multiple availability zones** Resilience  
Deploy CloudFormation across multiple availability zones. Use managed replicas or active-active configurations. Implement health checks and automatic failover. Test resilience regularly with chaos engineering or failover drills.

**A4 CloudFormation supports auto-scaling policies based on CPU, queue depth, or custom metrics.** Optimization  
CloudFormation supports auto-scaling policies based on CPU, queue depth, or custom metrics. Set minimum and maximum capacity. Horizontal scaling (more instances) is generally preferred over vertical for cloud workloads.

**A9 CI/CD pipelines use the cloud CLI or SDK to deploy updates to CloudFormation after tests pass.** Assessment  
CI/CD pipelines use the cloud CLI or SDK to deploy updates to CloudFormation after tests pass. Infrastructure-as-code (Terraform, CDK) defines CloudFormation configuration as version-controlled code deployed via the pipeline.

**A5 Place CloudFormation in a private subnet with no public endpoint where possible.** Federation  
Place CloudFormation in a private subnet with no public endpoint where possible. Use VPC security groups or firewall rules to allow only required traffic. Encrypt data at rest and in transit. Rotate credentials regularly.

**A10 Common mistakes include misconfigured IAM policies (too permissive), no cost alerting, deploying only in one availability zone, and missing backups.** Configuration  
Common mistakes include misconfigured IAM policies (too permissive), no cost alerting, deploying only in one availability zone, and missing backups. Read the service SLA carefully to understand what the cloud provider guarantees.